Chapter 10: Biomolecules

Reactions, Structures + Tips & Tricks

Formula Sheet

I. Carbohydrates – Structure-Establishing Reactions of Glucose

Reduction with HI (Confirms Straight Chain)

$$CH_2OH-(CHOH)_4-CHO \xrightarrow[\Delta]{HI} CH_3-(CH_2)_4-CH_3$$

Glucose reduced all the way to n-hexane confirms all six carbons are in a straight chain.

Reaction with Hydroxylamine (Confirms -CHO Group)

$$CH_2OH-(CHOH)_4-CHO + NH_2OH \longrightarrow CH_2OH-(CHOH)_4-CH{=}NOH + H_2O$$

Forms an oxime, confirming the presence of a carbonyl (aldehyde) group.

Reaction with HCN (Confirms -CHO, Chain Extension)

$$CH_2OH-(CHOH)_4-CHO + HCN \longrightarrow CH_2OH-(CHOH)_4-CH(OH)-CN$$

Forms a cyanohydrin; further confirms the carbonyl and is used in chain-extension proofs.

Bromine Water Test (Distinguishes Aldose from Ketose)

$$CH_2OH-(CHOH)_4-CHO \xrightarrow{Br_2\ \text{water}} CH_2OH-(CHOH)_4-COOH$$

Glucose (aldose) is oxidised by mild bromine water to gluconic acid; fructose (ketose) is *not* oxidised by bromine water – this is the standard test to distinguish the two.

Reaction with Dilute HNO_3 (Confirms Symmetry / 1° and 6° positions)

$$CH_2OH-(CHOH)_4-CHO \xrightarrow{\text{dil. } HNO_3} HOOC-(CHOH)_4-COOH$$

Both the -CHO and terminal -CH_2OH are oxidised to -COOH, forming saccharic acid; this dicarboxylic acid being optically inactive/meso-like helps establish the relative configuration.

Acetylation (Confirms Five -OH Groups)

$$CH_2OH-(CHOH)_4-CHO + 5\,(CH_3CO)_2O \longrightarrow CH_3COOCH_2-(CHOOCCH_3)_4-CHO$$

Forms glucose pentaacetate, confirming five -OH groups are present in the open-chain structure.

Osazone Formation (Phenylhydrazine, Confirms C1–C2)

$$CH_2OH-(CHOH)_4-CHO + 3\,C_6H_5NHNH_2 \longrightarrow CH_2OH-(CHOH)_3-C(=NNHC_6H_5)-CH=NNHC_6H_5 + C_6H_5NH_2 + NH_3 + 2\,H_2O$$

Excess phenylhydrazine reacts at C1 and C2 only, forming a bis-phenylhydrazone (osazone); glucose and fructose give the *same* osazone, proving they are identical from C3 to C6.

Cyclic Hemiacetal Formation

$$CH_2OH-(CHOH)_4-CHO \rightleftharpoons \alpha/\beta\text{-D-glucopyranose (cyclic hemiacetal)}$$

The -OH on C5 attacks the C1 carbonyl intramolecularly, forming a six-membered pyranose ring with a new -OH at C1 (anomeric carbon); explains mutarotation and why glucose does not give a Schiff's test.

II. Disaccharides – Glycosidic Linkage

Sucrose – Hydrolysis (Invert Sugar)

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{H^+ \text{ or invertase}} C_6H_{12}O_6\text{ (glucose)} + C_6H_{12}O_6\text{ (fructose)}$$

Sucrose is a non-reducing sugar: glycosidic linkage is between C1 of α-glucose and C2 of β-fructose, so both anomeric carbons are involved – no free -CHO/-C=O remains.

Maltose – Hydrolysis

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{H^+ \text{ or maltase}} 2\,C_6H_{12}O_6\text{ (glucose)}$$

Maltose is a reducing sugar: glycosidic linkage is α-1,4 between two glucose units, leaving one free anomeric -OH.

Lactose – Hydrolysis

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{H^+ \text{ or lactase}} C_6H_{12}O_6\text{ (glucose)} + C_6H_{12}O_6\text{ (galactose)}$$

Lactose (milk sugar) is a reducing sugar: β-1,4 glycosidic linkage between galactose and glucose, leaving one free anomeric -OH on the glucose unit.

III. Proteins – Amino Acids and Peptide Bond

Zwitterion Formation

$$H_2N-CHR-COOH \rightleftharpoons H_3N^+-CHR-COO^-$$

Amino acids exist predominantly as zwitterions (dipolar ions) in solution; the pH at which the molecule carries no net charge is the isoelectric point.

Peptide Bond Formation

$$H_2N-CHR-COOH + H_2N-CHR_2-COOH \xrightarrow{-H_2O} H_2N-CHR-CO-NH-CHR_2-COOH$$

Condensation between the -COOH of one amino acid and the -NH_2 of another forms a peptide (amide) bond, releasing water; chains of many such bonds form polypeptides/proteins.

IV. Reference Tables

Classification of Carbohydrates

Monosaccharides – cannot be hydrolysed further; classified as aldoses (aldehyde group, e.g., glucose) or ketoses (ketone group, e.g., fructose).
Oligosaccharides – 2–10 monosaccharide units on hydrolysis; disaccharides (sucrose, maltose, lactose) are the most common.
Polysaccharides – many monosaccharide units; starch and glycogen are energy-storage polymers of α-glucose, cellulose is a structural polymer of β-glucose. Polysaccharides are *not* sweet and are non-reducing.
Reducing vs. non-reducing: a sugar with a free anomeric -OH (free potential -CHO/C=O) is reducing (gives positive Fehling's/Tollens' test); one where both anomeric carbons are tied up in the glycosidic bond (e.g., sucrose) is non-reducing.

Protein Structure Levels

Primary – the linear sequence of amino acids joined by peptide bonds.
Secondary – regular folding patterns held by hydrogen bonds, e.g., α-helix and β-pleated sheet.
Tertiary – overall 3-D folding of a single polypeptide chain, giving the protein its specific biological activity.
Quaternary – spatial arrangement of two or more polypeptide subunits relative to each other (e.g., haemoglobin).
Denaturation – loss of secondary/tertiary/quaternary structure (biological activity lost) due to heat, pH change, etc., while the primary structure (peptide bonds) remains intact.

Vitamins – Classification and Deficiency

Fat-soluble: A (deficiency: night blindness/xerophthalmia), D (rickets), E (infertility), K (increased blood clotting time).
Water-soluble: B-complex and C – generally not stored in the body, so must be supplied regularly. B_1 deficiency: beriberi; B_2: cheilosis; B_6: convulsions; B_{12}: pernicious anaemia; C: scurvy.

Nucleic Acids – DNA vs. RNA

Basic unit is a **nucleotide** = nitrogenous base + pentose sugar + phosphate group. Bases are purines (adenine, guanine) and pyrimidines (cytosine, thymine/uracil).
DNA: sugar = deoxyribose; bases = A, T, G, C; double helical structure (Watson-Crick), strands held by H-bonding (A=T, G≡C); carries genetic information.
RNA: sugar = ribose; bases = A, U, G, C (uracil replaces thymine); usually single-stranded; helps in protein synthesis.

Tips, Tricks & Hacks

- **Glucose structure-proof chain, in order**: (1) HI reduction → straight 6-C chain; (2) reaction with HCN/NH_2OH → carbonyl present; (3) acetylation with 5 $(CH_3CO)_2O$ → 5 -OH groups; (4) bromine water oxidises it (not fructose) → the carbonyl is specifically an aldehyde (C1); (5) HNO_3 oxidises both ends to -COOH giving an optically active saccharic acid → fixes the open-chain structure; (6) same osazone as fructose → C3–C6 are identical in both sugars.
- **Why glucose doesn't fully answer to Schiff's/2,4-DNP tests despite having -CHO**: in solution it predominantly exists as the cyclic hemiacetal (pyranose form), which has no free aldehyde – this is also why it shows mutarotation (changing optical rotation as α and β anomers interconvert through the open-chain form).
- **Reducing sugar rule of thumb**: if the sugar can revert to an open-chain form with a free -CHO or adjacent -C(OH)=, it's reducing. All monosaccharides are reducing; among common disaccharides, only sucrose is non-reducing.
- **Isoelectric point**: below it, the amino acid carries a net positive charge (protonated -NH_3^+ dominates); above it, net negative (-COO^- dominates); exactly at it, the zwitterion has zero net charge and minimum solubility.
- **Denaturation vs. hydrolysis**: denaturation only disrupts secondary/tertiary/quaternary folding (physical), leaving peptide bonds (primary structure) intact; only chemical hydrolysis breaks the peptide bonds themselves.
- **DNA base-pairing memory aid**: "A-T, G-C" – Adenine pairs with Thymine (2 H-bonds), Guanine pairs with Cytosine (3 H-bonds) – G-C pairing is stronger.